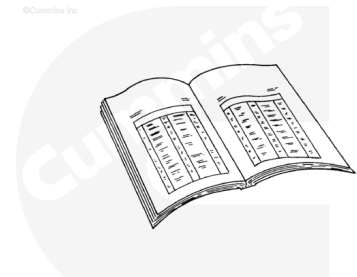


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



ck800wa

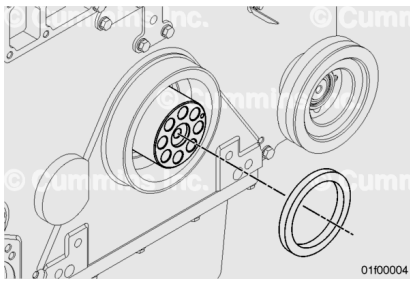
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Remove the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html>)
- Remove the vibration damper. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-052-tr.html>)
- Remove the crankshaft adapter. Refer to Procedure 001-017 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-017-tr.html>)

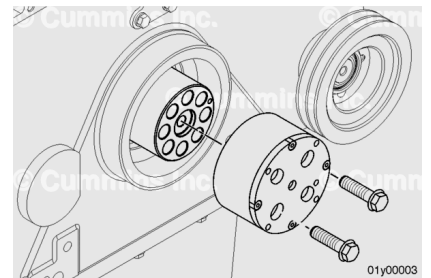
Remove

Remove the press-on shaft excluder (POSE) seal, if installed.



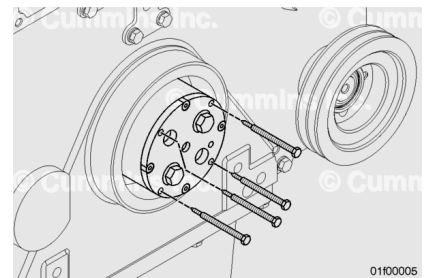
Use the crankshaft seal removal/installation tool, Part Number 5298874, to remove the front crankshaft seal.

Place the mandrel on the nose of the crank and secure using two of the special puller cap screws, Part Number 3163695, supplied with the tool.



Install the four self-drilling capscrews through the mandrel into the seal.

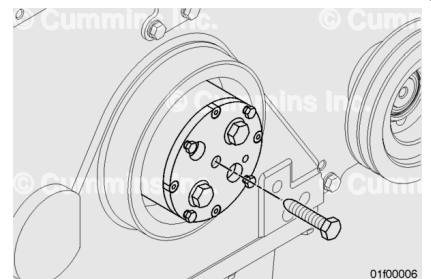
Install the self-drilling capscrews until approximately 5 mm [0.2 in] of the screw remains protruding from the mandrel.



Remove the securing capscrews.

Install the puller screw in to the mandrel to extract the seal from the gear housing.

Remove and discard the seal.

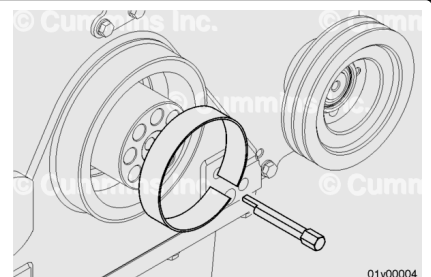


⚠ CAUTION ⚠

It is not necessary to completely cut the wear sleeve. If the sleeve is completely cut, the crankshaft can be damaged.

If a service seal with a wear sleeve has been installed, the wear sleeve **must** be removed from the crankshaft.

Install the protective sleeve, Part Number 3824764, in the seal bore of the front gear cover. The protective sleeve will protect the front gear cover bore from damage. Insert the expander



(chisel), Part Number 5298877, between the wear sleeve and protective sleeve.

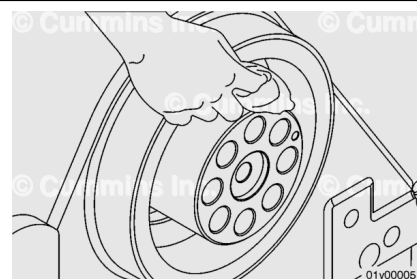
Turn the expander (chisel) so the nose deforms the sleeve. Repeat this procedure at three or four points around the sleeve.

The sleeve press fit will be reduced and can be lifted off the crankshaft. Remove the protective sleeve from the gear cover.

Clean

⚠ WARNING ⚠

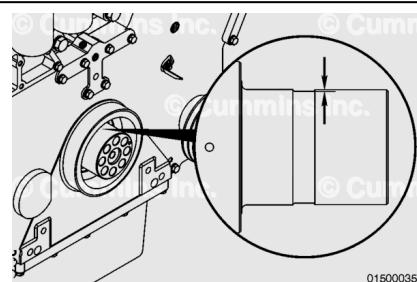
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the nose of the crankshaft and the bore in the front gear housing with a clean cloth and electrical contact cleaner, Part Number 3824510, or equivalent.

Check for damage on the seal area of the crankshaft.

If the crankshaft is grooved, a seal which contains a wear sleeve and special oversized seal **must** be installed. Refer to the Install section below for wear sleeve and seal assembly installation.

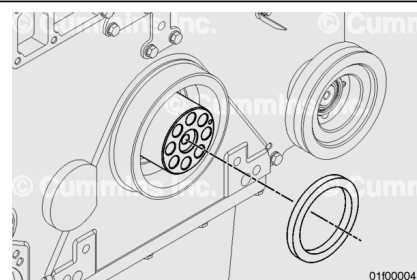


Install

⚠ CAUTION ⚠

Do not lubricate. The lips on the seal (1) and the seal surface of the crankshaft must be clean and dry.

⚠ CAUTION ⚠



Do not use a hammer on the tool. Damage to the threads will result. The tool can bind to the crankshaft because of the narrow tolerances.

⚠ CAUTION ⚠

Do not remove the seal from the wear sleeve. If the seal is removed from the wear sleeve, it can be damaged during the installation process.

⚠ CAUTION ⚠

The wear sleeve and the seal must be installed simultaneously using a special tool. Attempting to install the assembly without the tool will result in malfunction of the seal and an oil leak.

Standard Seal

- Use the installation sleeve that is included with the seal.
- Position the sleeve on the crankshaft.
- Push the seal (1) over the sleeve on the crankshaft.
- Push the seal until it touches the front cover.

Seal With Wear Sleeve

- Insert the seal and wear sleeve assembly onto the crankshaft.

Note : The front crankshaft seal with wear sleeve is recommended for use during rebuild or repair.

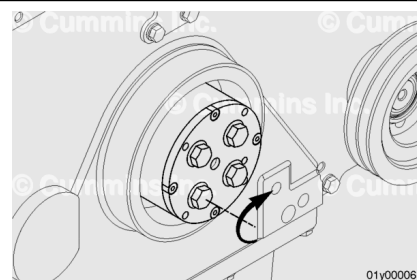
Press the seal into place using crankshaft seal replacer kit, Part Number 5298874.

Position the mandrel onto the crankshaft.

Install the four special puller capscrews, Part Number 3163695.

Tighten the capscrews alternately approximately ½ turn at a time until the tool touches the crankshaft nose.

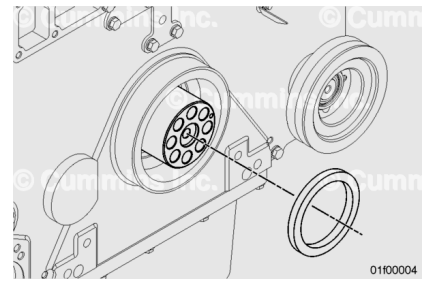
Remove the mandrel.



POSE

- Place the POSE seal onto the crankshaft.

- Push the seal on the crankshaft until it touches the seal.



Finishing Steps

⚠ WARNING ⚠

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⚠ WARNING ⚠

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- Install the crankshaft adapter. Refer to Procedure 001-017 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-017-tr.html>)
- Install the vibration damper. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-052-tr.html>)
- Install the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html>)
- Operate the engine and check for leaks.

